

# Lesson 59 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

**PYQ provenance:** Q1 and Q5 adapt Junior Assistant 2026 Q62; Q7 adapts Q64; Q11 and Q13 adapt Q65. The cited key is the official final key for JA 2026, and all adaptations are paraphrased, not verbatim.

- Q1. C.** von Neumann architecture stores instructions and data together in the same memory. A describes a separate-memory (Harvard-style) design; B and D wrongly keep only one of the two in memory.
- Q2. B.** Immediate addressing carries the operand inside the instruction encoding. A describes register addressing; C describes direct addressing; D confuses the operand with the next address.
- Q3. A.** The hit ratio is hits divided by total accesses. B gives the miss ratio; C and D invert the intended quantities.
- Q4. C.** A cache hit avoids a slow main-memory access because the cache is small, fast, and close to the CPU. A places the cache on secondary storage; B wrongly limits it to instructions; D gives it disk speed.
- Q5. D.** Hazards stem from dependencies between instructions, such as a read-after-write dependency exposed by pipelining. A wrongly restricts hazards to memory operands; B claims register addressing removes all hazards; C wrongly ties hazards to immediate addressing.
- Q6. D.** Cache is small fast memory between the CPU and main memory; virtual memory makes memory appear larger using secondary storage. A misplaces both; B treats them as one device; C reduces virtual memory to a single cache line.
- Q7. B.** The Program Counter holds the address of the next instruction to fetch. A is the standard trap (current instruction); C confuses it with an operand; D describes a past instruction.
- Q8. B.** The Instruction Register holds the instruction currently being fetched, decoded, and executed. A is the PC's role; C names an operand value; D names the data-bus width.
- Q9. D.** Interrupts are recognised at instruction boundaries during execution. A is the trap (only after the whole program); B and C wrongly limit interrupts to boot or to hardware faults.
- Q10. C.** Different instructions can take different numbers of clock cycles. A and B assert a constant count; D denies that instructions use cycles at all.
- Q11. D.** Overflow is detected when adding two same-sign operands yields an opposite-sign result. A is the trap (carry out of the MSB); B and C misstate the condition.

- Q12. B.** Zero-extension is for unsigned values and sign-extension is for signed values. A swaps them (the trap); C and D apply a single rule to both cases.
- Q13. D.** BCD encodes each decimal digit in 4 bits, so 25 is 0010 0101. A is the plain binary value; B uses eight bits per digit; C treats the number as hexadecimal.
- Q14. C.** Memory is byte-addressable, so each byte has its own address, and the effective address is the operand's actual location, which need not equal the instruction length. A and D wrongly equate unrelated quantities; B denies byte-addressability.
- Q15. A.** RISC is Reduced Instruction Set Computer and CISC is Complex Instruction Set Computer. B, C, and D give incorrect expansions of the abbreviations.
- Q16. D.** RISC uses simple, mostly fixed-length instructions, many registers, and a load/store architecture. A and C describe CISC traits; B contradicts the load/store discipline.
- Q17. B.** CISC offers a large set of complex, variable-length instructions and often permits memory operands. A and D describe RISC; C is false of CISC designs.
- Q18. C.** "All instructions take the same number of clock cycles" is false; the count varies. A, B, and D are all correct statements about the PC, interrupts, and overflow.
- Q19. A.** The BCD code for the digit 5 is 0101. B is the code for 9; C is the plain binary of 5, not its BCD code; D is not a valid BCD digit.
- Q20. A.** Statements I and II are true. III is false because overflow is a sign condition, not a carry out of the MSB. B includes false III; C includes III and omits II; D includes III.
- Q21. B.** The von Neumann bottleneck is the limited rate at which instructions and data move between the CPU and memory over a shared path. A, C, and D name unrelated delays or quantities.
- Q22. D.** Virtual memory makes memory appear larger than physical RAM by using secondary storage as backing store. A, B, and C describe unrelated or incorrect effects.
- Q23. A.** RISC means reduced, fast instructions. B, C, and D misstate the design idea.
- Q24. A.** CISC packs complex instructions. B, C, and D deny normal processor parts.
- Q25. A.** Intel Core chips continue the x86 CISC line. B, C, and D are non-processor objects.
- Q26. A, B, C.** Simple instructions, load-store style, and pipelining ease are correct. D is false: RISC still uses memory via loads/stores.
- Q27. A, B, C.** The instruction-count, code-density, and ARM facts are correct. D is false: every CPU has a control unit.
- Q28. C.** The Instruction Register holds the current instruction. MAR stores an address, PC the next instruction address, and the accumulator an intermediate result.
- Q29. A.** BCD encodes each decimal digit in four bits. It is not the same as converting the whole number to pure binary.

- Q30. A.** Zero extension pads an unsigned value with zeros. Sign extension copies the sign bit for signed values.
- Q31. A.** A data hazard is a dependency conflict between overlapping instructions. The other choices are peripheral or file-management issues.
- Q32. A, B, D.** PC, IR, and BCD statements are correct. Carry-out is not generally equivalent to signed overflow.

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